

✅ Experiment 1: Data Wrangling I

♦ Title: Data Wrangling using Python

♦ Objective

- To understand preprocessing of raw data
 - To clean and transform data before analysis
 - To prepare data for Machine Learning
-

♦ Theory

✅ What is Data Wrangling?

Data Wrangling (also called Data Munging) is the process of:

- Cleaning data
- Organizing data
- Transforming raw data into meaningful format

Real-world data is often:

- Incomplete
- Noisy
- Unstructured
- Missing values

Therefore, data wrangling is necessary before analysis or machine learning.

♦ Why Data Wrangling is Important?

1. Improves data quality
 2. Removes missing and inconsistent values
 3. Makes analysis easier
 4. Increases model accuracy
 5. Converts raw data into structured format
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♦ Steps of Data Wrangling

1. Discovering

Understand the dataset:

- Columns
 - Data types
 - Missing values
-

2. Structuring

Arrange data properly in rows and columns.

3. Cleaning

Remove:

- Null values
 - Duplicate values
 - Incorrect data
-

4. Enriching

Add useful information or transform features.

5. Validating

Check whether data is correct and consistent.

6. Publishing

Use cleaned data for:

- Analysis
 - Visualization
 - Machine Learning
-

◆ Important Concepts

✓ Missing Values

Missing values are empty values in dataset.

Example:

Name Age

Amit 20

Raj NaN

Functions Used:

isnull()
notnull()
fillna()
dropna()

 Data Formatting

Check correct datatype of columns.

Example:

- Age → integer
- Name → string

Function:

dtypes
astype()

 Data Normalization

Normalization means scaling data to a common range.

Example:

- Salary: 10000–50000
- Age: 18–60

Large values dominate ML models, so normalization is needed.

Types:

1. Min-Max Scaling
 2. Z-score Normalization
 3. Simple Feature Scaling
-

 Categorical to Numeric Conversion

Machine Learning works on numbers only.

Example:

Gender

Male

Female

Converted to:

Male Female

1 0

Methods:

1. Label Encoding
2. One Hot Encoding (get_dummies())

◆ Libraries Used

Library	Purpose
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pandas	Data handling
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numpy	Numerical operations
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matplotlib	Visualization
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◆ Dataset Used

Example Dataset:

Titanic / Cars / Adult dataset

Dataset contains:

- Numeric data
- Categorical data
- Missing values

◆ Algorithm / Procedure

Step 1:Import libraries

Step 2:Load dataset using pandas

Step 3:Display dataset

Step 4:Check missing values

Step 5:Get statistics using describe()

Step 6:Check dimensions using shape

Step 7:Check datatype using dtypes

Step 8:Convert datatype if needed

Step 9:Convert categorical variables into numeric

◆ **Explanation of Code**

✓ **import pandas as pd**

Imports pandas library for data analysis.

✓ **pd.read_csv()**

Loads CSV file into DataFrame.

✓ **head()**

Displays first 5 rows.

✓ **isnull()**

Checks missing values.

Returns:

- True → missing
 - False → not missing
-

✓ **describe()**

Gives statistics:

- Mean
 - Median
 - Standard deviation
 - Min
 - Max
-

✓ **shape**

Returns:

(rows, columns)

✓ size

Returns total number of elements.

✓ dtypes

Shows datatype of every column.

✓ astype()

Changes datatype.

Example:

```
df['Age'] = df['Age'].astype('int')
```

✓ get_dummies()

Converts categorical data into numeric.

Example:

Gender

Male

Female

becomes:

Male Female

1 0

0 1

◆ Expected Output

- Dataset loaded successfully
 - Missing values detected
 - Statistical summary displayed
 - Data types checked
 - Categorical variables converted into numeric
-

◆ Applications

1. Data preprocessing

2. Machine learning
 3. Data analysis
 4. Business intelligence
 5. Big data analytics
-

◆ Advantages

- Improves data quality
 - Removes inconsistencies
 - Better ML accuracy
 - Faster analysis
-

◆ Disadvantages

- Time-consuming
 - Large datasets need more processing
 - Wrong preprocessing can affect results
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◆ Conclusion

Data Wrangling is an important preprocessing step in Data Science. It converts raw and unstructured data into clean and meaningful data for analysis and machine learning.

◆ Frequently Asked Viva Questions

✓ 1. What is Data Wrangling?

Data wrangling is the process of cleaning and transforming raw data into usable format.

✓ 2. Why is Data Wrangling necessary?

Because real-world data contains:

- Missing values
- Errors
- Duplicate records
- Inconsistent formats

✓ **3. What is Data Preprocessing?**

It is the process of preparing data before analysis or machine learning.

✓ **4. What are missing values?**

Values that are empty or not available in dataset.

✓ **5. How do you detect missing values?**

Using:

isnull()

✓ **6. Difference between isnull() and notnull()?**

Function Purpose

isnull() Detect missing values

notnull() Detect non-missing values

✓ **7. What is normalization?**

Scaling data into common range.

✓ **8. Why convert categorical data into numeric?**

Machine learning algorithms work only on numerical data.

✓ **9. What is One Hot Encoding?**

Converting categorical data into binary columns.

✓ **10. What is Pandas?**

Python library used for data analysis and manipulation.

✓ **11. What is DataFrame?**

Two-dimensional tabular structure in pandas

✓ **12. Difference between shape and size?**

Shape

Size

Returns rows & columns Returns total elements

✓ **13. What does describe() do?**

Displays statistical summary of numeric columns.

✓ **14. What is the use of astype()?**

Used for datatype conversion

✓ **Experiment 2: Data Wrangling II (Detailed Viva Explanation)**

◆ **Title**

Data Wrangling II – Handling Missing Values, Outliers and Data Transformation

◆ **Objective**

- To understand handling of missing values
 - To detect and remove outliers
 - To perform data transformation and normalization
 - To prepare clean data for analysis and machine learning
-

◆ **Theory**

✓ **What are Missing Values?**

Missing values occur when data is not available.

Example:

Name Marks

Amit 85

Raj NaN

Here, NaN means missing value.

◆ **Causes of Missing Values**

1. Data entry mistakes

2. System errors
3. User skipped values
4. Data corruption

◆ Handling Missing Values

✅ 1. Ignore Missing Values

Remove rows containing missing values.

Function:

`dropna()`

✅ 2. Fill Missing Values

Replace missing values with:

- Mean
- Median
- Mode
- Constant value

Function:

`fillna()`

✅ 3. Replace Missing Values

Function:

`replace()`

◆ Important Functions

Function Purpose

`isnull()` Detect null values

`notnull()` Detect non-null values

`fillna()` Fill missing values

Function Purpose

dropna() Remove missing values

replace() Replace values

✅ Outliers

◆ What are Outliers?

Outliers are abnormal values that are very different from other values.

Example:

10, 12, 15, 14, 500

Here, 500 is an outlier.

◆ Why Outliers are Problematic?

- Reduce model accuracy
 - Distort statistical analysis
 - Affect mean and standard deviation
-

◆ Methods to Detect Outliers

✅ 1. Box Plot Method

Outliers appear outside whiskers.

✅ 2. Z-Score Method

Formula:

$$Z = \frac{x - \mu}{\sigma}$$
$$x$$
$$\mu$$
$$\sigma$$
$$z = \frac{x - \mu}{\sigma} \approx 1.2$$
$$\Phi(z) \approx 88.5\%$$

Where:

- x = data point
- μ = mean
- σ = standard deviation

If:

$$|Z| > 3$$

then value is considered outlier.

✓ 3. IQR Method (Most Important)

Formula:

$$IQR = Q_3 - Q_1$$

Where:

- Q_1 = 25th percentile
- Q_3 = 75th percentile

Outlier condition:

$$x < Q_1 - 1.5(IQR) \text{ or } x > Q_3 + 1.5(IQR)$$

◆ Data Transformation

✓ What is Data Transformation?

Converting data into another format for:

- Better understanding
 - Better model performance
 - Reducing skewness
-

◆ Types of Transformation

✓ 1. Normalization (Min-Max Scaling)

Converts data between 0 and 1.

Formula:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

✓ 2. Standardization (Z-score Scaling)

Transforms data with:

- Mean = 0
- Standard deviation = 1

Formula:

$$x' = \frac{x - \mu}{\sigma}$$

✓ 3. Robust Scaling

Uses:

- Median
- Interquartile Range

Useful for skewed data.

◆ Skewness

✓ What is Skewness?

Measures symmetry of data distribution.

Types:

Skewness Meaning

Positive Tail on right

Negative Tail on left

Zero Normal distribution

◆ Transformations to Reduce Skewness

Transformation	Use
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Log Transformation	Positive skew
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Square Root	Moderate skew
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Cube Root	Strong skew
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◆ Libraries Used

Library	Purpose
pandas	Data handling
numpy	Numerical operations
matplotlib	Visualization
seaborn	Boxplot
sklearn	Scaling

◆ Algorithm / Procedure

Step 1: Import libraries

Step 2: Create or load dataset

Step 3: Check missing values

Step 4: Handle missing values

Step 5: Detect outliers

Step 6: Remove outliers

Step 7: Apply data transformation

◆ Explanation of Code

✓ fillna()

Replaces missing values.

Example:

```
df['Age'].fillna(df['Age'].mean())
```

✓ dropna()

Removes rows with missing values.

✓ quantile()

Finds quartiles.

Example:

`Q1 = df['Marks'].quantile(0.25)`

✓ **Box Plot**

Used for outlier detection.

✓ **IQR Method**

Removes extreme values using quartiles.

✓ **Normalization**

Scales values between 0 and 1.

◆ **Expected Output**

- Missing values identified and filled
 - Outliers detected and removed
 - Data normalized successfully
-

◆ **Applications**

1. Data cleaning
 2. Data preprocessing
 3. Machine learning
 4. Statistical analysis
 5. Big data analytics
-

◆ **Advantages**

- Improves data quality
 - Better model accuracy
 - Removes noise from data
 - Helps achieve normal distribution
-

◆ **Disadvantages**

- Incorrect handling may lose important data
 - Outlier removal may affect results
-

◆ Conclusion

In this experiment, we handled missing values, detected outliers using IQR and boxplot, and applied data transformation techniques to improve data quality and prepare data for machine learning.

◆ Frequently Asked Viva Questions

✓ 1. What are missing values?

Values that are not present in dataset.

✓ 2. How do you detect missing values?

Using:

`isnull()`

✓ 3. Difference between `fillna()` and `dropna()`?

<code>fillna()</code>	<code>dropna()</code>
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Replaces missing values	Removes missing values
-------------------------	------------------------

✓ 4. What are outliers?

Extreme values different from normal data.

✓ 5. Why remove outliers?

Because they affect:

- Mean
 - Standard deviation
 - Model accuracy
-

✓ 6. What is IQR?

$$IQR = Q_3 - Q_1$$

Used to detect outliers.

✓ 7. What is normalization?

Scaling values between 0 and 1.

✓ 8. What is standardization?

Transforming data with mean = 0 and variance = 1.

✓ 9. What is skewness?

Measure of asymmetry in data distribution.

✓ 10. Which plot is used to detect outliers?

Boxplot.

✓ 11. What is Z-score?

$$Z = \frac{x - \mu}{\sigma}$$

$$x$$

$$\mu$$

$$\sigma$$

$$z = \frac{x - \mu}{\sigma} \approx 1.2$$

$$\Phi(z) \approx 88.5\%$$

Measures distance from mean.

✓ 12. Why is data transformation needed?

To improve model performance and reduce skewness.

✓ 13. What is the difference between normalization and standardization?

Normalization Standardization

Range 0 to 1 Mean 0, SD 1

✓ 14. What is quartile?

Division of data into four equal parts.

✓ 15. What is boxplot?

Graphical representation used to detect outliers.

✓ Experiment 3: Descriptive Statistics – Measures of Central Tendency and Variability

◆ Title

Descriptive Statistics using Python

◆ Aim / Problem Statement

Perform the following operations on any open-source dataset:

1. Provide summary statistics:
 - Mean
 - Median
 - Minimum
 - Maximum
 - Standard Deviation
 2. Group numeric data using categorical variables.
 3. Display statistical details of the Iris dataset species:
 - Iris-setosa
 - Iris-versicolor
 - Iris-virginica
-

◆ Objective

- To understand descriptive statistics
- To calculate measures of central tendency
- To analyze variability in data

- To use pandas statistical functions
-

◆ Theory

✓ What is Statistics?

Statistics is the science of:

- Collecting data
- Organizing data
- Analyzing data
- Interpreting data

It helps in decision making and prediction.

◆ Types of Statistics

Type	Meaning
Descriptive Statistics	Describes existing data
Inferential Statistics	Makes predictions from data

✓ Descriptive Statistics

Descriptive statistics summarize the dataset using numerical values.

Example:

- Average marks
 - Highest salary
 - Lowest age
-

◆ Measures of Central Tendency

Central tendency means finding the center value of data.

✓ 1. Mean (Average)

Mean is the average of all observations.

Formula:

$$\bar{x} = \frac{\sum x}{n}$$

Where:

- Σx = sum of all values
 - n = total observations
-

◆ Example

Values:

10, 20, 30

Mean:

$$\frac{10 + 20 + 30}{3} = 20$$

✓ 2. Median

Median is the middle value after sorting data.

◆ For Odd Number of Values

Example:

10, 20, 30

Median = 20

◆ For Even Number of Values

Example:

10, 20, 30, 40

Median:

$$\frac{20 + 30}{2} = 25$$

✓ 3. Mode

Mode is the most frequently occurring value.

Example:

10, 20, 20, 30

Mode = 20

◆ Measures of Variability / Dispersion

These show how spread out the data is.

✓ 1. Range

Difference between maximum and minimum value.

Formula:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

✓ 2. Standard Deviation

Measures deviation from mean.

Low SD → data close to mean

High SD → data widely spread

Formula:

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

◆ Important Statistical Functions

Function	Purpose
----------	---------

mean()	Average
--------	---------

median()	Middle value
----------	--------------

mode()	Frequent value
--------	----------------

std()	Standard deviation
-------	--------------------

min()	Minimum value
-------	---------------

max()	Maximum value
-------	---------------

describe()	Statistical summary
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◆ Grouping Data

✓ groupby()

Used to group data based on categorical columns.

Example:

```
df.groupby("gender")["age"].mean()
```

◆ Iris Dataset

The Iris dataset contains:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width
- Species

Species:

1. Iris-setosa
 2. Iris-versicolor
 3. Iris-virginica
-

◆ Libraries Used

Library Purpose

pandas Data handling

numpy Numerical calculations

◆ Algorithm / Procedure

Step 1:

Import libraries

Step 2:

Load dataset

Step 3:

Display dataset

Step 4:

Calculate statistics

Step 5:

Use groupby()

Step 6:

Analyze Iris species

◆ Explanation of Code

✓ `groupby()`

Groups data based on category.

Example:

```
df.groupby("gender")
```

✓ `mean()`

Calculates average.

✓ `median()`

Calculates middle value.

✓ `std()`

Calculates standard deviation.

✓ `describe()`

Displays:

- Count
 - Mean
 - Std
 - Min
 - Max
 - Percentiles
-

✓ `value_counts()`

Counts category frequency.

◆ Expected Output

- Statistical summary generated
 - Mean, median, SD calculated
 - Iris species analyzed successfully
-

◆ Applications

1. Data analysis
 2. Business analytics
 3. Machine learning preprocessing
 4. Financial analysis
 5. Medical statistics
-

◆ Advantages

- Easy understanding of data
 - Summarizes large datasets
 - Helps identify trends
-

◆ Disadvantages

- Does not predict future values
 - Cannot show complete data distribution
-

◆ Conclusion

In this experiment, descriptive statistical measures such as mean, median, mode, standard deviation, minimum, and maximum were calculated using Python and Pandas. Grouping operations were also performed using categorical variables.

◆ Frequently Asked Viva Questions

✓ 1. What is statistics?

Statistics is the science of collecting and analyzing data.

✓ 2. What is descriptive statistics?

It summarizes and describes data.

✓ 3. What is inferential statistics?

It predicts population behavior using sample data.

✓ 4. What is mean?

$$\bar{x} = \frac{\sum x}{n}$$

Average of observations.

✓ 5. What is median?

Middle value of sorted data.

✓ 6. What is mode?

Most frequently occurring value.

✓ 7. What is standard deviation?

Measure of spread of data from mean.

✓ 8. What is variance?

Square of standard deviation.

✓ 9. What is range?

$$Range = Maximum - Minimum$$

✓ 10. What does describe() do?

Displays statistical summary.

✓ 11. What is groupby() used for?

Grouping data based on categorical variables.

✓ 12. What is percentile?

Value below which a percentage of observations fall.

✓ 13. What is the Iris dataset?

Dataset containing flower measurements and species.

✓ 14. What is variability?

Spread of data values.

✓ 15. Difference between mean and median?

Mean

Median

Average

Middle value

Affected by outliers Not affected much

✓ 16. What is qualitative data?

Categorical data like gender or color.

✓ 17. What is quantitative data?

Numerical data like age or salary.